M4C 2025 - Assignment 2

Lan Stare

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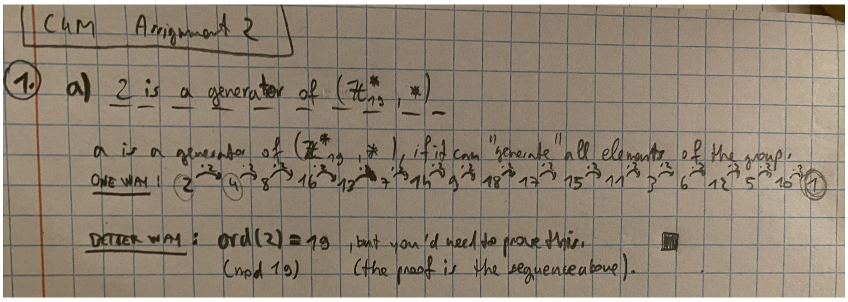
Problem 1

(a) 

C4M Assignment 2

1. a) 2 is a generator of $(\mathbb{Z}_{19}^*, *)$

a is a generator of $(\mathbb{Z}_{19}^*, *)$, if it can "generate" all elements of the group.

ONE WAY: $2 \xrightarrow{\cdot 2} 4 \xrightarrow{\cdot 2} 8 \xrightarrow{\cdot 2} 16 \xrightarrow{\cdot 2} 13 \xrightarrow{\cdot 2} 7 \xrightarrow{\cdot 2} 14 \xrightarrow{\cdot 2} 9 \xrightarrow{\cdot 2} 18 \xrightarrow{\cdot 2} 17 \xrightarrow{\cdot 2} 15 \xrightarrow{\cdot 2} 11 \xrightarrow{\cdot 2} 3 \xrightarrow{\cdot 2} 6 \xrightarrow{\cdot 2} 12 \xrightarrow{\cdot 2} 5 \xrightarrow{\cdot 2} 10 \xrightarrow{\cdot 2} 1$

BETTER WAY: $\operatorname{ord}(2) = 19 \pmod{19}$, but you'd need to prove this. (the proof is the sequence above). ∎

(b) Because 2 is a generator of the group $\mathbb{Z}_{19}$, every element is a power of 2 (mod 19). But if we're looking for an element which has order 18, we wouldn't find any, since the order of each element in our group has to be a devisor of 19. Since 18 and 19 are coprime, none of the elements can have the order 18.

(c) If a group G has a prime order p (p number of elements), then the order of every element in the group has to be either 1 or the prime number p. This is because of the corollary after the Langrange's theorem states, that the order of any element in the group has to devide the number of elements in that group. Since p is prime (it's only devisible by 1 and p), all of the orders are either 1 or p. Since 2 is the lowest prime number, we have 2 or more elements. If all the elements were of order 1, they would all be equivalent to the identity element, which would mean that we only had 1 element (contradictory with the fact that we have 2 or more elements). Therefore at least one element has order p, which means that it generates the whole group, which means the group is cyclic.

Problem 2

(a) ...

Problem 3

(a) ...